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Phosphine-catalyzed sequential (2+3)/(2+4) annulation of γ-vinyl allenoates: access to the synthesis of chromeno[4,3-*b*]pyrroles†‡

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A phosphine-catalyzed cascade (2+3)/(2+4) cyclization reaction of γ-vinyl allenoates with aldimine esters has been developed to provide a series of chromeno[4,3-*b*]pyrrole derivatives that contain three contiguous stereogenic centers. The method gives a good yield, excellent chemoselectivity and diastereoselectivity under mild conditions.

Chromeno[4,3-*b*]pyrrole and its structural analogues are ubiquitous moieties in natural products, biologically active products and pharmaceuticals (Fig. 1). For instance, lamellarins,[1] a large family of marine alkaloids isolated from diverse marine ascidians and sponges, have been identified as lead compounds for the inhibition of topoisomerase I and HIV-1 integrase. Martinelline or martinellic acid alkaloids[2] with a similar structural moiety of chromeno[4,3-*b*]pyrrole could act as a bradykinin receptor antagonist and have antibacterial activity. Chromeno[4,3-*b*]pyrrole derivatives also have promising utilities in medicinal chemistry, such as the treatment of Parkinson's disease, impulsive disorders and inflammation.[3]

Given their promising biological activities, chromeno[4,3-*b*]pyrrole frameworks have drawn considerable attention from both organic and medicinal chemists. To date, successful methodologies have been developed and the most accessible routes to chromeno[4,3-*b*]pyrrole skeletons involve 1,3-dipolar cycloaddition (1,3-DC) between alkenes and azomethine ylides as a key step.[4] In addition, organo-catalyzed, metal-free strategies have also completed the construction of chromeno[4,3-*b*]pyrrole frameworks in recent years.[5] Because of their remarkable potential pharmaceutical value, atom-economic, environment-friendly, and more efficient methods for the construction of the chromeno[4,3-*b*]pyrrole moiety with excellent yields and diastereoselectivity are still highly desired.

In 1995, the first nucleophilic phosphine-catalyzed annulation reaction of allenoates was reported by Lu.[6a] The seminal efficient methods for rapid construction of the five-membered carbocyclic moiety have been used in the total synthesis of natural products successfully.[6b,c] Since then, domino annulation reactions of unsubstituted allenoates catalyzed by nucleophilic phosphines have been one of the most efficient strategies for the rapid construction of cyclic products.[7] When there are different substituents at different sites in allenoates, they will exhibit diverse reactivity and could construct bicyclic compounds efficiently. For example, Huang's group[8] and other groups[9] realized phosphine-catalyzed bicyclic annulation reactions using γ-methyl or γ-benzyl allenoates, respectively. Moreover, the groups of Tong,[10] Shi[11] and Zhou[5e] have independently carried out interesting studies on allenoates with a leaving group such as an acetoxyl group at the β′ or δ-position to construct bicyclic skeletons. Recently, our group has designed δ-sulfonamido-allenoates and developed sequential (3+3)/(3+3) bicyclic annulation reactions using these new substituted allenoates successfully.[12]

The reactivity of substituted allenoates inspired us to install a γ-vinyl group on allenoates to extend the π-system and increase versatile reactivity. Fortunately, we first reported phosphine-catalyzed remote 1,7-addition for synthesis of diene carboxylates using γ-vinyl allenoates.[13] We assume that γ-vinyl allenoates could undergo sequential annulation reactions because of multiple reactive sites. In 2019, our group reported PPh_3-catalyzed (3+2)/(2+3) sequential annulation reactions

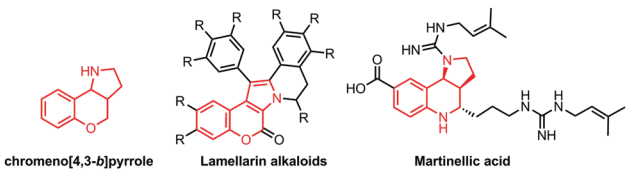


Fig. 1 Chromeno[4,3-*b*]pyrrole and natural products with an analogous core.

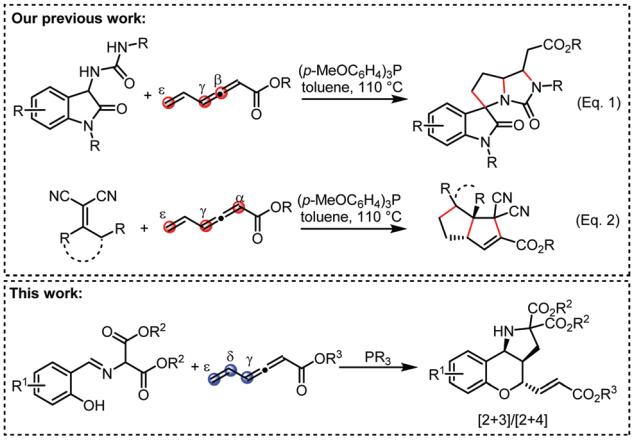
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† Dedicated to the 100th anniversary of Chemistry at Nankai University.

‡ Electronic supplementary information (ESI) available. CCDC 2075585 (**3aa**). For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/d1cc04199a

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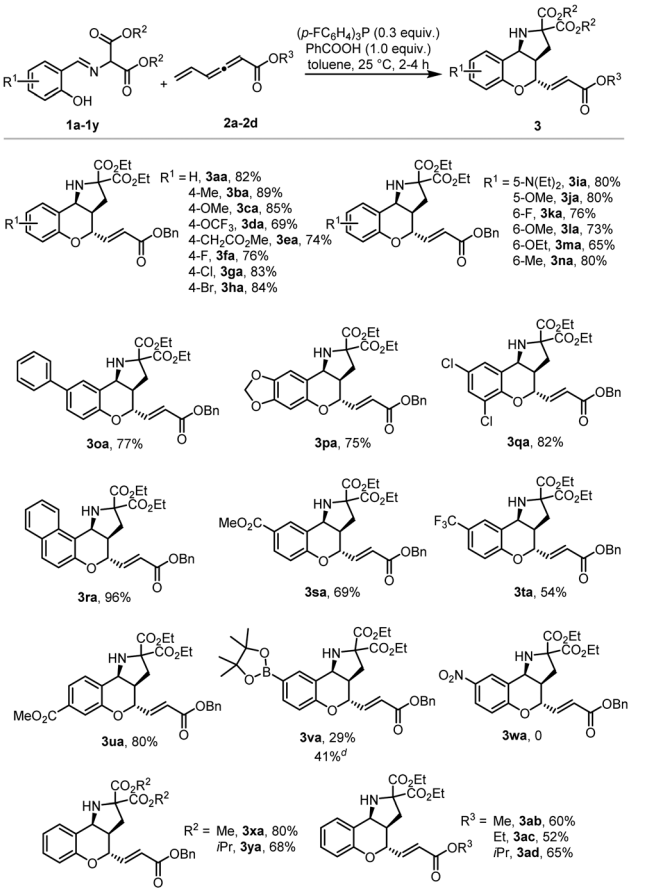
Scheme 1 Phosphine-catalyzed cascade annulation of γ-vinyl allenoates.

using γ-vinyl allenoates (eqn (1) in Scheme 1), in which they participated in reactions as a threefold preelectrophile.[14] Later, with our nuanced work on γ-vinyl allenoates, we successfully developed phosphine-catalyzed (3+2)/(3+2) annulation reactions using allenoates as a new reaction model to construct highly crowded fused carbocycles[15] (eqn (2) in Scheme 1). To investigate new reactive possibilities of γ-vinyl allenoates, herein, we report a phosphine-catalyzed reaction using aldimine esters as partners with three disparate reactive sites to develop new domino processes. However, several problematic issues need to be overcome: (1) chemoselectivity control of aldimine esters with one more nucleophile site; (2) regioselectivity control on reaction-sites of γ-vinyl allenoates, such as the electrophilic β position, and electrophilic γ or ε positions after nucleophilic phosphine addition forming a zwitterion; (3) difficult diastereoselectivity control of annulation products.

Initially, we started the investigation by employing aldimine ester (**1a**) and γ-vinyl allenoate (**2a**) as model substrates to validate the feasibility of our hypothesis. Gratifyingly, the desired product **3aa** in 50% yield was obtained catalyzed by PPh_3 (30 mol%) in toluene at 25 °C (see Table S1, entry 1, ESI†), and the structure and relative configuration of **3aa** was confirmed by X-ray crystallographic analysis. After a systematic screening of organophosphine catalysts was conducted, weakly nucleophilic phosphines (see Table S1, entries 1 and 2, ESI†) gave better results than strongly nucleophilic phosphines (see Table S1, entries 3–6, ESI†). To further improve the yield, additives were surveyed using $(p\text{-}FC_6H_4)_3P$ as the catalyst. These results revealed that weak acids had a positive impact and PhCOOH (100 mol%) was proved to be the best choice, affording **3aa** in 82% yield with excellent diastereoselectivity ratio (>20 : 1). Moreover, various solvents were investigated and the process gave the best outcome when using toluene as the solvent. Finally, using $(p\text{-}FC_6H_4)_3P$ (30 mol%) as the catalyst, and PhCOOH (100 mol%) as the additive and conducting the reaction in toluene at 25 °C for 2 hours were identified as optimal reaction conditions.

With the optimized reaction conditions in hand, the substrate scope and limitations of the reaction were surveyed and

Table 1 Reaction scope[abc]



[a] Unless otherwise noted, all the reactions were carried out under the optimal conditions. [b] Isolated yields. [c] All new products with an excellent dr (>20 : 1). [d] Added PhCOONa (1.0 equiv.) and PhCOOH (1.0 equiv.) as buffer.

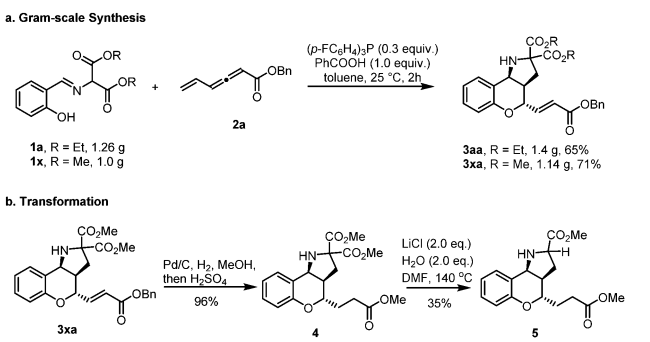
the results are shown in Table 1. The aldimine esters (**1b–1h**) bearing an electron-donating group or a halide as well as the ester-containing alkyl group (R^1 = 4-CH_2CO_2Me) at the *para* position of the phenol ring reacted smoothly and gave the corresponding products in good yields with excellent diastereoselectivity (**3ba–3ha** in Table 1). Similarly, the reaction of an aldimine ester bearing a monosubstituted group at the *ortho* and *meta* positions of the phenol ring also proceeded well and gave acceptable yields (**3ia–3na** in Table 1). The aldimine ester bearing an aromatic group at the *para* position of the phenol ring worked well and gave the expected compound **3oa** in 77% yield. It is worth noting that aldimine esters with a disubsitituted aromatic group went smoothly without an apparent decline in the yield (**3pa** and **3qa**). The aldimine ester bearing a naphthyl group proceeded very well, giving the product **3ra** in 96% yield. Furthermore, the reactivity of substrate aldimine esters with a strong electron-withdrawing group on the benzene ring was investigated. Most of these substrates could react smoothly in spite of the obvious electronic effects. Notably, the aldimine esters bearing $-CO_2Me$ or $-CF_3$ at the *para* position of the phenol ring led to moderate yields 69% and 54%,

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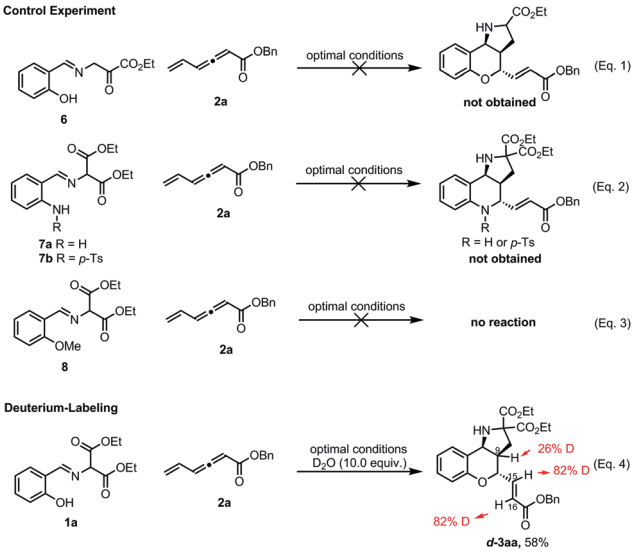
respectively (**3sa** and **3ta** in Table 1). In comparison, the reaction gave 80% (**3ua** in Table 1) yield when the $-CO_2Me$ group was at the *meta* position of the phenol ring. Because of strong electron-deficient trisubstituted boron atoms, the substrate **1v** gave a relatively low yield under the optimized reaction conditions. Interestingly, it could rise to 41% when the reaction was carried out with buffer (PhCOONa/PhCOOH, 1 equiv./1 equiv.). This may be due to the coordination effect of PhCOONa on boron, which reduces the electron deficiency of the boron atom and enhances the nucleophilicity of phenolic hydroxyl. However, with a nitro group at the *para* position of the phenol ring, the expected product could not be obtained (**3wa**). Furthermore, the effect of the ester substituent at the γ-vinyl allenoate (**2**) and malonate moiety was studied, and the desired products were obtained in moderate to good yields (**3xa**, **3ya**, **3ab**, **3ac**, and **3ad**).

To validate the practicality of this tandem annulation reaction, gram-scale experiments were carried out using **1a** (1.0 g, 3.6 mmol) and **1x** (1.0 g, 4.0 mmol), and obtained the corresponding products **3aa** (1.4 g, 65% yield) and **3xa** (1.14 g, 71%), respectively (Scheme 2a). To illustrate the potential utility of this new methodology, the post-functionalization of product **3xa** was carried out (Scheme 2b). The carbon–carbon double bond on the side chain of **3xa** was reduced by heterogeneous catalytic hydrogenation; meanwhile, the benzyl group was removed at the same time, followed by addition of catalytic amounts of sulfuric acid to afford the Fisher esterification product **4** in 96% yield in one pot. Next, **4** was decarboxylated by the Krapcho protocol[16] to give **5** in 35% yield.

Following the substrate scope evaluation, control experiments were carried out to gain insight into the mechanism of the reaction. Under the standard conditions, no corresponding product was obtained when allenoate **2a** reacted with ethyl (*E*)-2-[(2-hydroxybenzylidene)amino]acetate **6**, suggesting that the malonate moiety is necessary to trigger the reaction (eqn (1) in Scheme 3). Replacement of the hydroxyl group with an amido group or 4-methyl benzenesulfonamido group (*p*-TsNH) did not give the desired product, revealing that the hydroxyl group is essential (eqn (2) in Scheme 3). In addition, changing the hydroxyl group to a methoxy group, no new product was obtained (eqn (3) in Scheme 3). When 10 equiv. of D_2O was added to the reaction mixture under standard



Scheme 2 Gram-scale synthesis and transformation.



Scheme 3 Control and deuterium-labeling experiments.

conditions, we obtained the deuterated product **d-3aa** in 58% yield. ^{1}H NMR analysis showed remarkable H/D exchange at the C9, C15, and C16 positions.

Based on the results of control and deuterium-labeling experiments, the *ortho* hydroxyl group of the benzene ring was essential, and it might be important to increase the electrophilic reactivity of the imine *via* an intramolecular hydrogen bond. A putative mechanism of the cascade annulation reaction is illustrated in Fig. 2. First, the nucleophilic addition of $(p\text{-}FC_6H_4)_3P$ to allenoate **2a** followed by being protonated by PhCOOH led to the Michael acceptor **II**. Remote 1,7-addition[13] of the reactive enolate **I** to the acceptor **II** formed the first σ-bond and generated intermediate **III**. The intermediate **III** could resonate to give **IV**, which was activated by an intramolecular hydrogen bond and carried out intramolecular addition of the carbon anion to vinyl phosphonium to afford intermediate **V**. Then after the addition of phenolate to the electron deficient double bond, the second cyclization was done *via* intramolecular formal (2+4) cycloaddition. Finally,

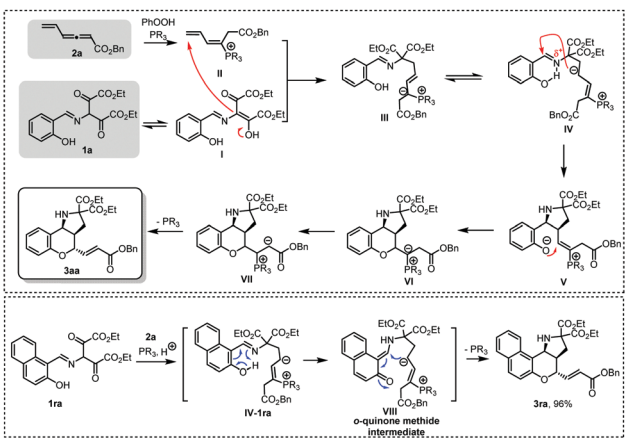


Fig. 2 Plausible catalytic cycle.

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the elimination of the phosphine catalyst yielded product **3aa**, and regenerated the phosphine catalyst $(p\text{-}FC_6H_4)_3P$, which continued the next catalytic cycle.

It should be noted that the naphthyl-substituted substrate **1ra** gave the target product in almost quantitative yield (**3ra** in Table 1). We assumed that there was another mechanism (bottom in Fig. 2). The resonated intermediate **IV-1ra** after remote 1,7-addition went through [1,5] H-shift and dearomatization to obtain the highly active *o*-quinone methide intermediate **VIII**, which then carried out the subsequent reaction to give the target compound **3ra**. A similar mechanism was also proposed by many groups for addition reactions with *o*-quinone methide as a key active intermediate.[17] Extension of the aromatic system in substrate **1ra** allowed for a decreased energetic penalty for the breaking of aromaticity in the phenyl core, and formed the *o*-quinone methide intermediate efficiently during the reaction, and other substrates had not these intrinsic qualities. This may be the reason for the high yield of the product **3ra**.

In conclusion, we have developed a phosphine-catalyzed cascade (2+3)/(2+4) annulation reaction of γ-vinyl allenoates with aldimine esters. This procedure constructs three σ-bonds and three contiguous stereogenic centers in a one-pot method to assemble chromeno[4,3-*b*]pyrrole derivatives directly in good yields with excellent chemoselectivity and diastereoselectivity under mild conditions. Finally, control and deuterium-labeling experiments provided deeper insight into the reaction mechanism. Mechanistic studies revealed that an intramolecular hydrogen bond is important for the reaction to run smoothly.

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Conflicts of interest

There are no conflicts to declare.

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